Q1.Design and implement a stack using an array without using any built-in stack library.
Perform the following operations:

- PUSH(x)
- POP()
- PEEK()
- DISPLAY()

Your program must handle both **Stack Overflow** and **Stack Underflow** conditions.

Additional Task:
Explain the **time complexity and space complexity** of each operation. Also discuss what happens when the stack size is fixed and the user attempts to insert more elements than its capacity.

Q2. **Implement a Circular Queue using an array.** The queue should support:

- ENQUEUE(x)
- DEQUEUE()
- FRONT()
- DISPLAY()

The implementation must correctly distinguish between a **full queue and an empty queue**.

Additional Task:
Compare the circular queue with a simple linear queue and explain:

1. Why a circular queue provides better utilization of memory.
2. Time complexity of ENQUEUE and DEQUEUE.
3. Space complexity of the queue.
4. What problem occurs in a linear queue when REAR reaches the last index even though unused positions exist at the beginning?